

A Queueing-Network Simulator for an LLM Inference Serving Pipeline

Phase-3 final presentation: what the validated simulator revealed.

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Why model LLM serving as a queueing network?

- Requests wait at admission and tokenization.
- Requests wait for batching.
- Requests wait for GPU service.

the scheduler knobs (B, τ, c)

These knobs change where requests wait.

A validated simulator lets us study these trade-offs reproducibly.

The three-stage queueing model

Q1, admission and tokenization.

Geo/D/1, one server, deterministic service.

$$S_1 = 2 \text{ cycles}$$

Q2, batching scheduler.

FCFS; a batch is released when B requests have collected, or when the oldest has waited τ cycles. $B = 1$: pass-through.

$$S_{\text{batch}}(b) = \lceil S_3 + \delta (b - 1) \rceil$$

$$\rho_3 = \frac{\lambda_{\text{batch}} E[S_{\text{batch}}]}{c} < 1$$

Q3, GPU workers.

c parallel workers; one worker serves one batch, all its requests share the duration.

$$B = 1: \rho_3 = \lambda_3 E[S_3]/c < 1 \quad \lambda_{\text{batch}}: \text{Q2}$$

batch-dispatch rate. Single-server nodes:

$$\rho_i = \lambda_i E[S_i] < 1.$$

Bernoulli(λ) arrivals; one cycle = one scheduler tick.

Simulator and cycle semantics

1. **Arrival** Bernoulli(λ) request joins the queue
2. **Service countdown** busy workers count down or complete
3. **Completion and routing** finished work moves downstream
4. **Dispatch** idle servers take the head of the line
5. **Counters** occupancy, delay, utilization update

One scheduler tick = one simulation cycle: fixed-increment time advance, the course's cycle-driven scheme. Same-cycle immediate access: a request arriving at cycle t can enter service at cycle t .

Validation against analytical queueing models

$$\rho = \lambda S \qquad L_q = \lambda W_q$$

$$W_q = \frac{\rho (S - 1)}{2 (1 - \rho)} \qquad L_q^{\text{Geo/Geo/1}} = \frac{2\lambda^2}{1 - 2\lambda}$$

Anchors: single-station configurations of the engine are exactly our validated single-queue simulators.

The W_q expression is the general one for our slotted convention; extending the sweep to $S = 3$ showed that the L_q form we had initially carried over is **specific to $S = 2$** .

Criterion: worst relative error below 1% at 10^7 cycles: $\lambda \leq 0.45$ for $S = 2$, $\lambda \leq 0.30$ for $S = 3$; lengths $10^3, 10^5, 10^7$.

Validation results

Geo/D/1, $S = 2$: **0.75%**

Geo/D/1, $S = 3$: **0.46%**

Geo/Geo/1, mean 2: **0.94%**

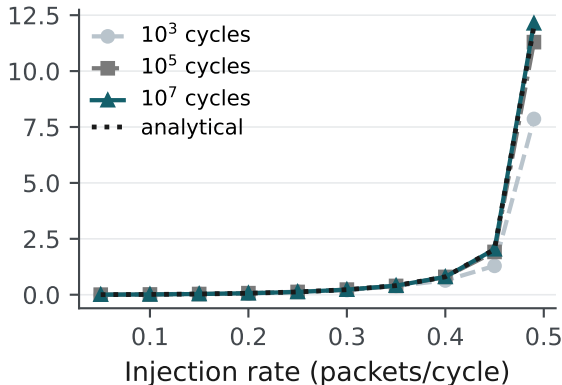
Factor-2 ratio: **0.95%** from 2

Network accounting: 3.4×10^{-7} .

Utilization and throughput vs. load: 0.11%.

$W = L_q/\lambda$ per node: machine precision.

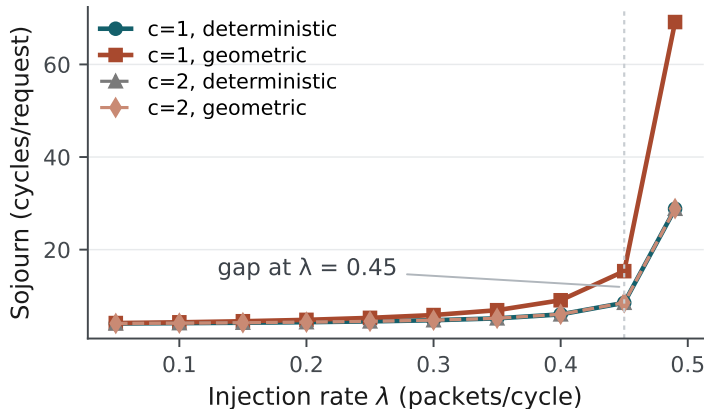
$\lambda = 0.49$: informational.



Convergence onto the analytical curve:

7.6% $\rightarrow$ 1.8% $\rightarrow$ 0.46% at $\lambda = 0.3$.

E2: service variability under load



At $\lambda = 0.45$, $c = 1$:

- deterministic: 8.519 cycles
- geometric: 15.350 cycles
- increase: +6.831 cycles (80.2%)

The premium grows with load: 2.1% at $\lambda = 0.05$, 139.9% at $\lambda = 0.49$.

Service variability becomes costly as load approaches saturation.

E2: parallelism and a structural result

det., $c = 1$: 8.519

geo., $c = 1$: 15.350

geo., $c = 2$: 8.549

$\Delta = 0.03$ vs det.

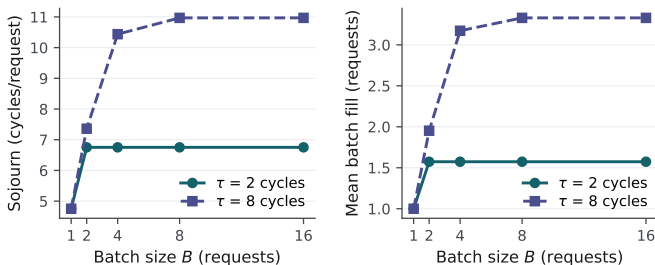
$\lambda = 0.45$; paired arrival streams.

Structural result. With deterministic $S_1 = S_3 = 2$ and pass-through Q2, Q1 departures are at least two cycles apart, so Q3 never accumulates a queue: $L_{q,Q3} = 0$ in all 22 deterministic rows.

Tested configuration only.

E3: the batching trade-off

$$\lambda = 0.3, \quad \delta = 0.5$$
$$B \in \{1, 2, 4, 8, 16\}, \quad \tau \in \{2, 8\}$$



Larger batches improve GPU utilization, but Q2 waiting dominates at this load. $B = 1$: utilization 0.60, Q2 wait 0.75 cycles $\rightarrow B = 8, \tau = 8$: utilization 0.31, wait 5.39 cycles; batch service 2.0 $\rightarrow$ 3.4 cycles. Length robustness: 10^6 vs 10^7 runs, 0.004 cycles (0.29%).

E3: saturation and interpretation

$B : 4 \rightarrow 8 \rightarrow 16$

fill at $\tau = 8 : 3.17 \rightarrow 3.33 \rightarrow 3.33$

Empirical saturation knee between $B = 4$ and $B = 8$ at $\tau = 8$.

Larger batches improve batch fill, lower GPU utilization (0.60 $\rightarrow$ 0.31), but lengthen Q2 waiting (0.75 $\rightarrow$ 5.39 cycles) and raise end-to-end delay (4.752 $\rightarrow$ 10.969 cycles); throughput stays within 0.06% of λ .

In the tested grid, $B = 1$ has the lowest end-to-end delay.

Key takeaways

1. **Validated simulator.** Single-queue analytical checks within 1% at converged loads; network accounting to 3.4×10^{-7} .
2. **Variability and parallelism shape latency.** Geometric GPU service increases delay strongly at one worker; a second worker largely absorbs it.
3. **Batching trades delay for efficiency.** Larger batches improve utilization but increase waiting; saturation near $B = 8$ for $\tau = 8$; $B = 1$ delays least in the tested grid.

Limitations: linear feedforward chain; single Bernoulli input; static batching; batch-level service abstraction; two service laws; tick-quantized time; cycle mapping not calibrated. Network-level analytical validation remains open pending course material.

Thank you.